ARM-450 — Kinematics, Workspace and Hardware Feasibility

- Robot: ARM-450, 6-DOF, 450 mm overall, 360 mm horizontal reach
- Convention: Modified (Craig) DH + velocity-propagation Jacobian — the user's own reference notation
- Date: 2026-08-19
- Source of truth: arm450.urdf, generated from the CAD parameters. Where the DH model and the URDF disagree, the URDF is correct.

1. Convention

Frame i is attached to link i. Parameters are (α_{i−1}, a_{i−1}, d_i, θ_i):

- a_{i−1} (link length) — mutual-perpendicular distance between the two Z axes, measured along X_{i−1}
- α_{i−1} (link twist) — angle from Z_{i−1} to Z_i, measured about X_{i−1}
- d_i (link offset) — distance from X_{i−1} to X_i, measured along Z_i
- θ_i (joint angle) — angle from X_{i−1} to X_i, measured about Z_i

The Modified-DH homogeneous transform:

$$
{}^{i-1}_{i}T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1} d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

2. DH table

Verified against arm450.urdf to 0.000 nm over 5,000 random poses.

i	α_{i−1}	a_{i−1} (mm)	d_i (mm)	θ_i	joint
1	0°	0	90	q_1	base yaw
2	−90°	0	0	$q_2 - 90°$	shoulder pitch
3	0°	119	0	$q_3 - 90°$	elbow pitch

i	α_{i−1}	a_{i−1} (mm)	d_i (mm)	θ_i	joint
4	−90°	0	181	q_4	forearm roll
5	+90°	0	0	q_5	wrist pitch
6	−90°	0	0	q_6	tool roll

Tool offset: d_{tool} = 60 mm along Z_6.

Where the numbers come from:

- d_1 = 90 = base height 50 + shoulder rise 40
- a_2 = 119 = the upper-arm link, J2 → J3
- d_4 = 181 = forearm shell 119 + J4→J5 offset 62
- d_{tool} = 60 = J5→J6 30 + J6→TCP 30

A caution worth recording: my first hand-derived twist column was wrong and produced a 650 mm position error. A constrained search over α ∈ {0, ±90°} and θ-offset ∈ {0, ±90°} found the table above. Always verify a DH table numerically against the URDF — do not trust a hand derivation.

3. Forward kinematics

$$ {}^{0}_{6}T(q) = {}^{0}_{1}T\,{}^{1}_{2}T\,{}^{2}_{3}T\,{}^{3}_{4}T\,{}^{4}_{5}T\,{}^{5}_{6}T $$

$$ {}^{0}_{TCP}T = {}^{0}_{6}T \cdot Trans_z(d_{tool}), \qquad d_{tool} = 60\ mm $$

Verification

check	result
Max position difference, DH vs URDF (5,000 poses)	0.0000 µm
Rotation difference	constant 180° about Z — a tool-frame labelling offset only
Home pose q = 0	TCP at (0, 0, 450) mm
q_2 = 90°	TCP at (360, 0, 90) mm

The 180° rotation is a fixed relabelling of the tool frame's X/Y axes. Position is unaffected; applying it makes the two models identical in orientation as well.

4. Transform matrices at the home pose

With all q = 0:

```
0_1T = [ 1  0  0  0     ]      1_2T = [ 0  1  0  0     ]
       [ 0  1  0  0     ]             [ 0  0  1  0     ]
       [ 0  0  1  0.090 ]             [ 1  0  0  0     ]
       [ 0  0  0  1     ]             [ 0  0  0  1     ]

2_3T = [ 0  1  0  0.119 ]      0_6T = [-1  0  0  0     ]
       [-1  0  0  0     ]             [ 0 -1  0  0     ]
       [ 0  0  1  0     ]             [ 0  0  1  0.390 ]
       [ 0  0  0  1     ]             [ 0  0  0  1     ]
```

TCP = 0_6T · Trans_z(0.060) → (0, 0, 0.450) m, which is the 450 mm requirement met exactly.

5. Jacobian by velocity propagation

Propagating frame by frame from base to tip:

$$ {}^{i+1}\omega_{i+1} = {}^{i+1}_{i}R\,{}^{i}\omega_i + \dot{\theta}_{i+1}\,{}^{i+1}\hat{Z}_{i+1} $$

$$ {}^{i+1}v_{i+1} = {}^{i+1}_{i}R\left({}^{i}v_i + {}^{i}\omega_i \times {}^{i}P_{i+1}\right) + \dot{d}_{i+1}\,{}^{i+1}\hat{Z}_{i+1} $$

For a revolute joint the $\dot{d}$ term drops; for a prismatic joint the θ term drops. Expressed in the base frame, each column becomes:

$$ J_i = \begin{bmatrix} \hat{Z}_i \times (P_{tcp} - P_i) \\ \hat{Z}_i \end{bmatrix} $$

Verification: compared against numerical differentiation of the FK over 400 random poses — max column error 1.85×10^{-8} m/rad, which is finite-difference noise.

6. Real workspace

Computed from the verified DH model over the true joint limits, 250,000 samples.

$$ R_{max} = a_2 + d_4 + d_{tool} = 119 + 181 + 60 = 360\ mm $$

quantity	value
Chain length, fully extended	450.0 mm
Max distance from the base origin	450.0 mm

quantity	value
Max horizontal reach from the J1 axis	360.0 mm
Min horizontal radius (dead zone)	0.3 mm
Height range	−200 … +450 mm
Poses below the mounting plane	23.4 %
Reachable volume (10 mm voxels)	118 litres
Fill fraction of a 360 mm sphere	60.4 %

The dead zone is effectively zero

$$r_{inner} = \max\left(0,\ |a_2 - d_4| - d_{tool}\right) = \max(0,\ 62 - 60) = 2\ mm\ \rightarrow\ measured\ 0.3\ mm$$

The naive two-link formula $|a_2 - d_4|$ = 62 mm over-reports, because it ignores that the 60 mm wrist can point inward. The measured minimum radius over 250k poses is 0.3 mm.

Task-constrained workspace — the number that matters

Reachability alone is not enough; the pen must also be normal to the surface.

On a grid of 260 points over a vertical board:

	result
Drawable with the pen normal	136/260 = 52.3 %
Best radial band	x = 240 … 300 mm (13–15 of 20 heights)
Comparison: myCobot 280 (thesis)	57.9 %

ARM-450 is essentially as capable as the myCobot for vertical surface tracing.

7. Hardware feasibility

Sine tracing — the immediate task

Run through Rsine's own DrawingPlane and sine_waypoints, imported unchanged from the frozen package with only the kinematics swapped:

board distance	solved	position rms	pen tilt
200 mm	25/40	9.09 mm	31.9°
240 mm	40/40	0.028 mm	0.5°
280 mm	39/40	0.445 mm	3.1°
300 mm	40/40	0.007 mm	0.2°

Verdict: feasible. Mount the board at 240–300 mm and the arm traces the sine with the pen effectively perpendicular.

Horizontal table — not feasible as designed

0/40 at every height tested. J2, J4 and J5 reach their limits simultaneously. Widening J5 from ±110° to ±150° cuts the pen-tilt error from 54° to 16° but still does not close it. This is a regression against the myCobot, which manages 63.5 % on horizontal.

Actuation

joint	worst-case torque	ST3215 (30 kgf·cm)	verdict
J1	0.000 N·m	—	gravity cannot load a vertical yaw axis
J2	2.625 N·m	89 % of stall	too much to hold with 300 g at full extension
J3	1.271 N·m	43 %	marginal
J4, J5	0.296 N·m	10 %	fine
J6	0.088 N·m	3 %	fine

With a pen instead of a 300 g payload, J2 falls to 14–29 % of stall across the drawing band. A uniform ST3215 at every joint is therefore adequate for the sine-tracing task — and all six joints now physically accept the same servo.

Accuracy budget

source	contribution
Servo backlash, 0.5° over three joints	4.82 mm
Joint bearing clearance, preloaded pair at 22 mm spacing	0.65 mm
Unbolted clamshell seam (torsion)	2.80 mm
Elastic droop, PLA+CF	0.11 mm

Backlash dominates. No amount of structural stiffening improves it — that requires either better servos or joint-side encoders.

8. Verdict

Feasible for the immediate task, with three conditions:

1. Vertical board at 240–300 mm. Horizontal-table tracing does not work without wider J2/J5 travel or a wrist redesign.
2. Pen-scale payload. The 300 g rating at full extension exceeds what an ST3215 holds continuously at J2.
3. Expect ~5 mm absolute accuracy from backlash. Adequate for drawing; not adequate for the MASCOT docking precision that follows.

The kinematics, the 450 mm envelope, the zero dead zone and the 52.3 % task workspace are all confirmed. What remains unproven is everything mechanical: the 0.65

mm wobble figure is calculated and never measured, and four of the six joint parts are first-pass with no fit check against real hardware.

Modified (Craig) DH — derived, and cross-checked three ways

Added 2026-08-25 at the user's request: build the DH table and the IK in their convention — modified/Craig DH with a velocity-propagation Jacobian — then check it against ikpy, an outside package.

The table

i	α_{i−1}	a_{i−1} (mm)	d_i (mm)	θ_i
1	0°	0	90	q_1 + 180°
2	90°	0	0	q_2 + 90°
3	0°	119	0	q_3 + 90°
4	90°	0	181	q_4 + 180°
5	90°	0	0	q_5 + 180°
6	90°	0	0	q_6

Tool: 60 mm along Z_6 to the TCP.

How it was obtained, and the two numbers that are not what they look like

The joint axes were taken from the URDF at q = 0 and the common normals computed between consecutive axes:

```
J1  origin (0, 0,  50)   axis Z         Z1 ⊥ Z2   a = 0
J2  origin (0, 0,  90)   axis Y         Z2  Z3   a = 119   <- the only non-zero a
J3  origin (0, 0, 209)   axis Y         Z3 ⊥ Z4   a = 0
J4  origin (0, 0, 328)   axis Z         Z4 ⊥ Z5   a = 0
J5  origin (0, 0, 390)   axis Y         Z5 ⊥ Z6   a = 0
J6  origin (0, 0, 420)   axis Z
```

Every axis passes through the base Z line, so every a is zero except the one between the two parallel pitch axes.

d_4 = 181, not 119. This is the trap. Frame 3's origin sits where X_3 meets Z_3 (z = 209) and frame 4's sits where Z_4 meets Z_5 (z = 390), so the offset along Z_4 is 119 + 62. Writing 119 — the "obvious" link length — puts the tool 193 mm out, which is exactly what the first version of this file did.

a_2 = 119, not d_3. Parallel axes have a real common perpendicular, and it goes in a.

The Jacobian

Built by velocity propagation, base to tip, per the reference method:

```
i+1_ω_{i+1} = i+1_i R · i_ω_i  +  θ̇_{i+1} · Ẑ
i+1_v_{i+1} = i+1_i R · ( i_v_i + i_ω_i × i_P_{i+1} )
```

not by finite differencing. The prismatic branch is kept in the code even though ARM-450 is all-revolute, because the reference method covers both.

Cross-check — three implementations sharing no code

check	result
FK: Craig DH == URDF chain	1.96×10^{-13} mm over 200 random poses
FK: Craig DH == ikpy	2.17×10^{-13} mm
FK: URDF chain == ikpy	1.14×10^{-13} mm
tool axis agrees	8.95×10^{-16}
Jacobian: propagation == finite difference	1.81×10^{-5}
IK (Craig DH + propagated Jacobian)	median 1.6×10^{-11} mm
IK (ikpy)	median 0.000 mm

7 of 7 pass. Agreement between the first two proves the DH table; agreement with ikpy proves the first two do not share a mistake — which is the failure a self-written cross-check cannot catch.

> ikpy does not import cleanly on this machine and the reason is worth recording. sympy stringifies numpy scalars; numpy 2 changed their repr to np.float64(1.0); sympy then cannot parse its own input. ikpy converts numpy scalars internally, so it dies before doing any kinematics. check_ikpy.py routes the converter through float() instead of the string path. Nothing to do with the arm — but a check that cannot run is not a check, and it would have been easy to report "ikpy unavailable" and move on.

Run it: python3 check_ikpy.py